3.33 Complete Discrete Valuation Rings

We isolate the algebraic (topology-free) notion of completeness for a DVR, needed to make sense of "formal" local models such as $k[[t]]$, and prove the key structural fact used throughout: a complete DVR has a **unique** extension of its valuation to any finite separable extension of its fraction field.

This generalises the subsection on Extensions of Power Series Rings above, which proved the case $\mathcal{O}_K \cong k[[t]]$ by direct computation in power series, without this general completion machinery: (3.32.35) and (3.32.36) there are the special cases of (3.33.19) and (3.33.20) here.

Definition 3.33.1 (Complete DVR)
A DVR A with maximal ideal $\mathfrak{m}$ is complete *if every sequence $(\overline{y}_n)_{n\geq 0}$ with $\overline{y}_n \in A/\mathfrak{m}^{n+1}$, compatible under reduction, is the image of some element of A (equivalently, $A \to \varprojlim_n A/\mathfrak{m}^{n+1}$ is surjective; see (3.33.5) for the general criterion).*

Proposition 3.33.2 (The Formal Power Series Ring is a Complete DVR)
Let k be a field. Then $k[[t]]$ is a complete DVR.

Proof. Drafted by Claude Sonnet, reviewed by David Rufino.

By (3.32.16), $k[[t]]$ is a discrete valuation ring, so it remains to show completeness.

Let $(\overline{y}_n)_{n\geq 0}$ with $\overline{y}_n \in k[[t]]/(t)^{n+1}$ be compatible under reduction. Each $\overline{y}_n$ has a unique representative $f_n = \sum_{i\leq n} c_i^{(n)} t^i$ of degree at most n. Compatibility says $f_n \equiv f_{n-1} \pmod{t^n}$, so $c_i^{(n)} = c_i^{(n-1)}$ for $i < n$. Hence for each i the coefficient $c_i := c_i^{(n)}$ is independent of $n \geq i$. Put $f := \sum_i c_i t^i \in k[[t]]$. For each n, the coefficients of f up to degree n are those of f_n, so $f \equiv f_n \pmod{t^{n+1}}$, that is $\overline{y}_n$ is the reduction of f. Thus $k[[t]]$ is complete. ∎

Proposition 3.33.3 (A Complete DVR with a Coefficient Field is a Power Series Ring)
Let $(A, \mathfrak{m})$ be a complete DVR, π a uniformiser, and $k \subseteq A$ a coefficient field ($k(\mathfrak{m}) = k$). Then there is a k-algebra isomorphism $k[[Y]] \xrightarrow{\sim} A$, $Y \mapsto \pi$.

Proof. Drafted by Claude Sonnet, reviewed by David Rufino.

Coefficients. Since $A/\mathfrak{m} = k$ and $\mathfrak{m} = (\pi)$, every $a \in A$ can be written $a \equiv \sum_{i<N} c_i \pi^i \pmod{\pi^N}$ with $c_i \in k$, and the c_i are uniquely determined by a (apply v). Indeed $a - c_0 \in \mathfrak{m} = \pi A$ gives $a = c_0 + \pi a_1$, and iterate.

Definition of Φ. Let $f = \sum_n c_n Y^n \in k[[Y]]$ with truncations $s_N := \sum_{n<N} c_n \pi^n$. Since $s_{N+1} - s_N = c_N \pi^N \in \pi^N A$, the sequence $(s_N \mod \pi^N)_N$ is compatible under reduction, so by completeness of A (3.33.1) there is $\Phi(f) \in A$ with $\Phi(f) \equiv s_N \pmod{\pi^N}$ for every N. It is unique, as $\bigcap_N \pi^N A = 0$.

Homomorphism. Φ is k-linear, and $\Phi(1) = 1$. For f, g, the truncations satisfy $s_N(f)s_N(g) \equiv s_N(fg) \pmod{\pi^N}$ (the products agree modulo π^N, since the difference only involves coefficients of degree $\geq N$ in Y, sent to multiples of π^N). Hence $\Phi(f)\Phi(g) \equiv s_N(f)s_N(g) \equiv s_N(fg) \equiv \Phi(fg) \pmod{\pi^N}$ for all N, so $\Phi(f)\Phi(g) = \Phi(fg)$. Additivity is the same argument.

Injective. If $\Phi(f) = 0$ and $f \neq 0$, let n_0 be minimal with $c_{n_0} \neq 0$. Then $\pi^{n_0} c_{n_0} \equiv \Phi(f) - \sum_{n>n_0} c_n \pi^n \equiv 0 \pmod{\pi^{n_0+1}}$, so $c_{n_0}\pi^{n_0} \in \pi^{n_0+1}A$, contradicting the uniqueness of the coefficients above.

Surjective. Given $a \in A$, take its coefficients $c_i \in k$ from the first paragraph. These are compatible (the expansion modulo π^N is the truncation of the expansion modulo π^{N+1}), so $f := \sum_i c_i Y^i$ satisfies $\Phi(f) \equiv a \pmod{\pi^N}$ for every N, whence $\Phi(f) = a$. ∎

Definition 3.33.4 (Completion of a Local Ring)
Let $(A, \mathfrak{m})$ be a local ring, with $\mathfrak{m}$-adic filtration $\mathfrak{m}_\bullet$ (3.13.4) (so $\mathfrak{m}_n = \mathfrak{m}^{n+1}$). Write $\hat{A} := \hat{A}_\mathfrak{m}$ for its $\mathfrak{m}$-adic completion, $\iota := \iota_\mathfrak{m} : A \to \hat{A}$ for the canonical map, and $\lambda_n : \hat{A} \to A/\mathfrak{m}^{n+1}$ for its projections (3.13.19).

For $x = (\overline{a}_n)_{n\geq 0} \in \hat{A}$ and $j \geq k \geq 0$: $\overline{a}_j \equiv \overline{a}_k \pmod{\mathfrak{m}^{k+1}}$ (repeated compatibility (3.13.16)).

Proposition 3.33.5 (Completeness Criterion)
Let $(A, \mathfrak{m})$ be a local ring with $\bigcap_n \mathfrak{m}^n = 0$ (automatic if A is a DVR (3.30.9)), and $\hat{A}$, ι as in (3.33.4). Then the following are equivalent:

a) $\iota : A \to \hat{A}$ is surjective;
b) every sequence $(\overline{a}_n)_{n\geq 0}$, $\overline{a}_n \in A/\mathfrak{m}^{n+1}$, compatible under reduction ($\overline{a}_n \equiv \overline{a}_{n-1} \bmod \mathfrak{m}^n$ for $n \geq 1$), equals $(a \bmod \mathfrak{m}^{n+1})_n$ for some $a \in A$;
c) for some strictly increasing sequence $n_0 < n_1 < n_2 < \cdots$ in $\mathbb{N}$ (equivalently, for every such sequence), every sequence $(\overline{a}_N)_{N\geq 0}$, $\overline{a}_N \in A/\mathfrak{m}^{n_N+1}$, compatible under reduction, equals $(a \bmod \mathfrak{m}^{n_N+1})_N$ for some $a \in A$.

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*When these hold, ι is an isomorphism, and we say A is **complete**.*

Proof. Drafted by Claude Sonnet, reviewed by David Rufino.

This is the special case $\mathfrak{a} := \mathfrak{m}$ of (3.13.21), noting $\hat{A}_{\mathfrak{m}} = \hat{A}$, $\iota_{\mathfrak{m}} = \iota$. ∎

Proposition 3.33.6 (The Completion of a Local Ring is Local)
Let $(A, \mathfrak{m})$ be a local ring, and $\hat{A}$, λ_n as in (3.33.4). Then $\hat{A}$ is local, with units, for $(a_n)_{n\geq 0} \in \hat{A}$, characterised by

$$(a_n)_{n\geq 0} \in \hat{A}^\star \iff a_0 \in (A/\mathfrak{m})^\star$$

maximal ideal $\hat{\mathfrak{m}} := \ker \lambda_0 = \hat{A} \setminus \hat{A}^\star$, characterised, for $(a_n)_{n\geq 0} \in \hat{A}$, by

$$(a_n)_{n\geq 0} \in \hat{\mathfrak{m}} \iff a_0 = 0$$

and λ_0 inducing an isomorphism $\hat{A}/\hat{\mathfrak{m}} \xrightarrow{\sim} A/\mathfrak{m}$: that is, $\hat{A}$ and A have the same residue field.

Proof. **Units.** One direction is obvious by applying λ_0. Conversely suppose $x = (a_n)_{n\geq 0} \in \hat{A}$ and $a_0 \in (A/\mathfrak{m})^\star$ (so in particular $a_0 \neq 0$ (3.3.7)). Then by definition $\rho_{n,0}(a_n) = a_0$ which by (3.3.56) means $a_n \notin \mathfrak{m}/\mathfrak{m}^{n+1}$. As $A/\mathfrak{m}^{n+1}$ is local with maximal ideal $\mathfrak{m}/\mathfrak{m}^{n+1}$ (3.19.4), $a_n \notin \mathfrak{m}/\mathfrak{m}^{n+1}$ gives $a_n \in (A/\mathfrak{m}^{n+1})^\star$ (3.19.3). Let $b_n := (a_n)^{-1}$, then we claim the b_n are compatible. For $n \geq 1$: $1 = \rho_{n,n-1}(a_n b_n) = \rho_{n,n-1}(b_n)\rho_{n,n-1}(a_n) = \rho_{n,n-1}(b_n)a_{n-1}$. By uniqueness of inverses (3.1.5), $\rho_{n,n-1}(b_n) = b_{n-1}$. Therefore these determine a well-defined element $y = (b_n)_{n\geq 0} \in \hat{A}$ which is an inverse for x as required.

Locality. $\ker \lambda_0$ is an additive subgroup of $\hat{A}$ (3.3.14), so by (3.19.3) it is enough to show that $\ker \lambda_0 = \hat{A} \setminus \hat{A}^\star$ which will also give the form of the maximal ideal $\hat{\mathfrak{m}}$. However $A/\mathfrak{m}$ is a field (3.3.59) so by the first part

$$(a_n)_{n\geq 0} \in \ker \lambda_0 \iff a_0 = 0 \iff a_0 \notin (A/\mathfrak{m})^\star \iff (a_n)_{n\geq 0} \notin \hat{A}^\star$$

that is, $\ker \lambda_0 = \hat{A} \setminus \hat{A}^\star$, which is also the unique maximal ideal.

Residue field. As λ_0 is surjective (3.13.24) with $\ker \lambda_0 = \hat{\mathfrak{m}}$, (3.3.57) gives the claimed isomorphism $\hat{A}/\hat{\mathfrak{m}} \xrightarrow{\sim} A/\mathfrak{m}$. ∎

Lemma 3.33.7 (Finite Free Modules over a Complete Ring are Complete)
Let $(A, \mathfrak{m})$ be a complete local ring (3.33.5), and M a free A-module of finite rank r. Then

$$\iota_M : M \to \varprojlim_{N\geq 1} M/\mathfrak{m}^N M, \quad x \mapsto (x \mod \mathfrak{m}^N M)_{N\geq 1},$$

is a bijection.

Proof. Drafted by Claude Sonnet, reviewed by David Rufino.

Fix an A-basis $e_1, \ldots, e_r$ of M, giving an A-module isomorphism $\phi : A^r \xrightarrow{\sim} M$, $(a_i)_{i=1}^r \mapsto \sum_i a_i e_i$; note $\mathfrak{m}^N M = \mathfrak{m}^N\left(\sum_i Ae_i\right) = \sum_i \mathfrak{m}^N e_i$ for every $N \geq 1$.

Existence of $\overline{\phi}_N$. Fix $N \geq 1$, and write $\sigma_N : A^r \to (A/\mathfrak{m}^N)^r$ for the coordinatewise reduction (a surjection with $\ker \sigma_N = (\mathfrak{m}^N)^r$) and $\pi_N : M \to M/\mathfrak{m}^N M$ for the quotient map. We first check $\ker(\pi_N \circ \phi) = (\mathfrak{m}^N)^r$ too: for $(a_i)_i \in A^r$, $\phi\big((a_i)_i\big) = \sum_i a_i e_i$, so $(a_i)_i \in \ker(\pi_N \circ \phi) \iff \sum_i a_i e_i \in \mathfrak{m}^N M = \sum_i \mathfrak{m}^N e_i \iff a_i \in \mathfrak{m}^N$ for every i (the forward implication using injectivity of ϕ: if $\sum_i a_i e_i = \sum_i c_i e_i$ for $c_i \in \mathfrak{m}^N$, then $a_i = c_i \in \mathfrak{m}^N$ for every i), that is, $(a_i)_i \in (\mathfrak{m}^N)^r = \ker \sigma_N$.

In particular $\ker \sigma_N \subseteq \ker(\pi_N \circ \phi)$, so by the universal property of the quotient σ_N (3.3.101), there is a unique A-module homomorphism $\overline{\phi}_N$ making

$$\begin{array}{ccc} A^r & \xrightarrow{\quad\phi\quad}_{\sim} & M \\ {\scriptstyle\sigma_N}\big\downarrow & & \big\downarrow{\scriptstyle\pi_N} \\ (A/\mathfrak{m}^N)^r & \xrightarrow[\overline{\phi}_N]{} & M/\mathfrak{m}^N M \end{array}$$

commute, that is $\overline{\phi}_N \circ \sigma_N = \pi_N \circ \phi$, explicitly $\overline{\phi}_N\big((\overline{a}_i)_{i=1}^r\big) = \sum_i a_i e_i \mod \mathfrak{m}^N M$ for any representatives $a_i \in A$ of $\overline{a}_i$.

$\overline{\phi}_N$ is an isomorphism: by (3.3.101), $\ker\big(\overline{\phi}_N\big) = \ker(\pi_N \circ \phi)/\ker \sigma_N$, and this is 0 as $\ker(\pi_N \circ \phi) = \ker \sigma_N$ (shown above), so $\overline{\phi}_N$ is injective; it is also surjective, as $\pi_N \circ \phi = \overline{\phi}_N \circ \sigma_N$ is (a composite of the surjections ϕ, π_N) and σ_N is surjective.

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Compatibility of the $\overline{\phi}_N$. For $P \geq N \geq 1$, the square

$$\begin{array}{ccc} (A/\mathfrak{m}^P)^r & \xrightarrow[\sim]{\overline{\phi}_P} & M/\mathfrak{m}^P M \\ \downarrow & & \downarrow \\ (A/\mathfrak{m}^N)^r & \xrightarrow[\overline{\phi}_N]{\sim} & M/\mathfrak{m}^N M \end{array}$$

(vertical arrows the evident reduction maps (3.3.56)) commutes: both composites send the class of $(a_i)_i \in A^r$ to $\sum_i a_i e_i \mod \mathfrak{m}^N M$, directly from the explicit formula for $\overline{\phi}_N$ above (resp. for $\overline{\phi}_P$, followed by reduction mod $\mathfrak{m}^N M$). Thus $\left(\overline{\phi}_N\right)_{N\geq 1}$ is a compatible family of isomorphisms between the towers $\left((A/\mathfrak{m}^N)^r\right)_{N\geq 1}$ and $(M/\mathfrak{m}^N M)_{N\geq 1}$.

Passing to the limit. By (3.3.157) (applied to r copies of the tower $(A/\mathfrak{m}^N)_{N\geq 1}$), $\prod_{i=1}^r \varprojlim_N A/\mathfrak{m}^N \cong \varprojlim_N (A/\mathfrak{m}^N)^r$; composing with $\varprojlim \overline{\phi}$ (an isomorphism, (3.3.156)) gives an isomorphism

$$\left(\varprojlim_N A/\mathfrak{m}^N\right)^r \xrightarrow{\sim} \varprojlim_N M/\mathfrak{m}^N M$$

sending $\left((a_i \mod \mathfrak{m}^N)_N\right)_{i=1}^r$ to $\left(\sum_i a_i e_i \mod \mathfrak{m}^N M\right)_N$ for $a_i \in A$ (by the explicit formula for $\overline{\phi}_N$ above), that is, under the fixed basis, ι_M corresponds to $\iota_A^{\times r}$ (the r-fold product of $A \to \varprojlim_N A/\mathfrak{m}^N$, $a \mapsto (a \mod \mathfrak{m}^N)_N$). As A is complete, $\iota_A : A \to \hat{A} = \varprojlim_n A/\mathfrak{m}^{n+1}$ is bijective (3.33.5) reindexing ($N := n+1$), $A \to \varprojlim_N A/\mathfrak{m}^N$, $a \mapsto (a \mod \mathfrak{m}^N)_N$, is bijective too, hence so is its r-fold product, hence so is ι_M. ∎

Proposition 3.33.8 (The Completion is a DVR)
Let $(A, \mathfrak{m})$ be a DVR, $K := \mathrm{Frac}(A)$, and $\hat{A}$ the completion (3.33.4), $K_{\mathfrak{m}} := \mathrm{Frac}\left(\hat{A}\right)$. Then $\hat{A}$ is a DVR. Fixing a uniformiser π of A and writing $\hat{\pi} := \iota(\pi)$:

a) $\hat{\pi}$ is a uniformiser of $\hat{A}$;
b) the valuation of $\hat{A}$ is given, for $0 \neq x = (\overline{a}_m)_{m\geq 0} \in \hat{A}$, by $v_{\hat{A}}(x) = \min\{m \geq 0 : \overline{a}_m \neq 0\}$;
c) the maximal ideal of $\hat{A}$ is $\hat{\mathfrak{m}} = (\hat{\pi}) = \left\{(\overline{a}_m)_{m\geq 0} \in \hat{A} : \overline{a}_0 = 0\right\} = \ker \lambda_0$;
d) the units of $\hat{A}$ are $\hat{A}^\star = \left\{(\overline{a}_m)_{m\geq 0} \in \hat{A} : \overline{a}_0 \neq 0\right\}$;
e) $\ker\left(\hat{A} \to A/\mathfrak{m}^{n+1}\right) = \hat{\mathfrak{m}}^{n+1}$, for every $n \geq 0$ (generalising c), the case $n = 0$).

Consequently, the valuations v of A and $v_{\hat{A}}$ of $\hat{A}$ extend uniquely (3.30.2) to $K^\star$ and $K_{\mathfrak{m}}^\star$ respectively, explicitly by

$$v(a/b) := v(a) - v(b) \quad (a, b \in A \setminus \{0\}), \qquad v_{\hat{A}}(x/y) := v_{\hat{A}}(x) - v_{\hat{A}}(y) \quad \left(x, y \in \hat{A} \setminus \{0\}\right).$$

Proof. **Units**: By (3.33.6) $\hat{A}$ is local with unique maximal ideal $\hat{\mathfrak{m}} := \ker \lambda_0$; so $\hat{A}^\star = \hat{A} \setminus \ker \lambda_0 = \left\{(\overline{a}_m)_{m\geq 0} \in \hat{A} : \overline{a}_0 \neq 0\right\}$.

$\hat{\pi}$ is a non-unit; $\widehat{A}$ is not a field. As ι is injective and $\pi \neq 0$, $\hat{\pi} = \iota(\pi) \neq 0$. Writing $\hat{\pi} = (\overline{\pi}_m)_m$, $\overline{\pi}_0 = \pi \mod \mathfrak{m} = 0$ (as $\pi \in \mathfrak{m}$), so $\hat{\pi} \notin \hat{A}^\star$ by d). As every non-unit of a local ring lies in its maximal ideal (3.19.3), $\hat{\pi} \in \ker \lambda_0$; so $\hat{\mathfrak{m}} := \ker \lambda_0 \neq 0$ (as $\hat{\pi} \neq 0$), and $\hat{A}$ is therefore not a field (a field has no nonzero proper ideal).

Valuation of $\widehat{A}$. Define

$$v_{\hat{A}}\left((a_n)_{n\geq 0}\right) := \min\{m \mid a_m \neq 0\}$$

For $j \geq k$ we may show that $a_j \in \mathfrak{m}^k \setminus \mathfrak{m}^{k+1}$ and therefore $v(a_j) = k$ (3.30.3).

$\widehat{A}$ is a domain. Suppose $0 \neq x = (\overline{a}_m)_m$, $0 \neq y = \left(\overline{b}_m\right)_m \in \hat{A}$ with $xy = 0$; set $k := v_{\hat{A}}(x)$, $l := v_{\hat{A}}(y)$. Representatives $a, b \in A$ of $\overline{a}_{k+l}, \overline{b}_{k+l}$ have $v(a) = k$, $v(b) = l$ so $v(ab) = k + l$ (3.30.2), that is $ab \notin \mathfrak{m}^{k+l+1}$ (3.30.3) but ab represents the $(k+l)$-th coordinate of $xy = 0$, forcing $ab \in \mathfrak{m}^{k+l+1}$: a contradiction. So $\hat{A}$ is a domain.

$\widehat{A}$ is a DVR. We show that for every $0 \neq x$ there is a $w \in \hat{A}^\star$ and $k \geq 0$ such that $x = w\hat{\pi}^k$, namely criteria (3.30.2) e) to be a DVR.

Suppose $x = (\overline{a}_n)_{n\geq 0}$ with some representatives $a_n \in A$. Let $k := v_{\hat{A}}(x)$ and recall $j \geq k \implies v(a_j) = k$, so by the same criteria $a_j = u_j \pi^k$ for some $u_j \in A^\star$ uniquely depending on a_j.

Define $\overline{w}_n := u_{n+k} + \mathfrak{m}^{n+1}$. By definition of a_n,

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$$a_{n+k+1} \equiv a_{n+k} \mod \pi^{n+k+1} \Longrightarrow u_{n+k+1} \equiv u_{n+k} \mod \pi^{n+1}$$

by cancelling π^k. This means precisely that $\rho_{n+1,n}(\overline{w}_{n+1}) = \overline{w}_n$ so $w := (\overline{w}_n)_{n\geq 0}$ is a well-defined element of $\hat{A}$.

Recall $\overline{w}_0 = u_k + \mathfrak{m}$. By assumption $u_k \in A^\star$, whence $u_k \notin \mathfrak{m}$, which shows $\overline{w}_0 \neq 0$. Therefore by Units, we have $w := (\overline{w}_n)_{n\geq 0} \in \hat{A}^\star$.

By definition $\lambda_n\big(w\hat{\pi}^k\big) = \overline{w}_n\big(\pi^k + \mathfrak{m}^{n+1}\big) = u_{n+k}\pi^k + \mathfrak{m}^{n+1} = a_{n+k} + \mathfrak{m}^{n+1} = \overline{a}_n = \lambda_n(x)$. By construction of $\hat{A}_n$ as an inverse limit (3.3.154) we have $x = w\hat{\pi}^k$ as required.

Valuation, maximal ideal, kernel of λ_n. For $0 \neq x = (\overline{a}_m)_m \in \hat{A}$, the uniformised representation gives $x = \hat{\pi}^k u$ for the unit u constructed above and $k = \min\{m \geq 0 : \overline{a}_m \neq 0\}$; by the uniqueness of such a representation (3.30.2), $v_{\hat{A}}(x) = k$, giving b).

Let $x = (a_n)_{n\geq 0}$ then $\lambda_n(x) = 0 \iff a_n = 0 \iff a_0 = \ldots = a_n = 0 \iff v_{\hat{A}}(x) \geq n+1$. Then using that $\hat{A}$ is a DVR with unique maximal ideal $\hat{\mathfrak{m}} := (\hat{\pi})$ this is equivalent to $x \in \hat{\mathfrak{m}}^{n+1}$ (3.30.3) giving e); of which c) is a special case. ∎

Lemma 3.33.9 (Local is Continuous, DVR Case)
Let $(A, \mathfrak{m})$, $(B, \mathfrak{n})$ be DVRs, and $\phi : A \to B$ a ring homomorphism. Then ϕ is local (that is $\phi(\mathfrak{m}) \subseteq \mathfrak{n}$) if and only if ϕ is continuous (3.13.5) and then $\phi(\mathfrak{m}^{n+1}) \subseteq \mathfrak{n}^{n+1}$ for every $n \geq 0$ (same-index witness).

Proof. Drafted by Claude Sonnet, reviewed by David Rufino.

This is (3.13.15) with $\mathfrak{a} := \mathfrak{m}$, $\mathfrak{b} := \mathfrak{n}$ ($\mathfrak{n}$ maximal, hence radical (3.3.36)); the same-index witness $\phi(\mathfrak{m}^{n+1}) \subseteq \mathfrak{n}^{n+1}$ is exactly that lemma's forward-direction computation ($k := n+1$). ∎

Proposition 3.33.10 (Taut Criterion, DVR Case)
Let $(A, \mathfrak{m})$, $(B, \mathfrak{n})$ be DVRs, $\phi : A \to B$ a local ring homomorphism, and e its ramification index (3.30.10). Then ϕ is taut (3.13.10) if and only if $e = 1$ and $B = \phi(A) + \mathfrak{n}$.

Proof. Drafted by Claude Sonnet, reviewed by David Rufino.

Fix a uniformiser π of A, so $e = v_B(\phi(\pi))$ (3.30.10). For every $n \geq 0$: $\phi(\mathfrak{m}^{n+1}) = \phi((\pi^{n+1})) = (\phi(\pi)^{n+1})$, which is $\mathfrak{n}^{e(n+1)}$ if $e < \infty$ (as $v_B(\phi(\pi)^{n+1}) = e(n+1)$), and 0 if $e = \infty$.

Case $e = \infty$. For any strictly increasing $n_0 < n_1 < n_2 < \cdots$, the taut recursion at N reads $\mathfrak{n}^{n_N+1} = \phi(\mathfrak{m}^{n_N+1}) + \mathfrak{n}^{n_{N+1}+1} = \mathfrak{n}^{n_{N+1}+1}$, forcing $n_N + 1 = n_{N+1} + 1$, that is $n_N = n_{N+1}$: impossible, as $n_N < n_{N+1}$. So ϕ is not taut.

Case $e < \infty$. As ideals of a DVR are totally ordered by inclusion, $\mathfrak{n}^{e(n_N+1)} + \mathfrak{n}^{n_{N+1}+1} = \mathfrak{n}^{\min(e(n_N+1), n_{N+1}+1)}$; so the taut recursion at N, $\mathfrak{n}^{n_N+1} = \mathfrak{n}^{\min(e(n_N+1), n_{N+1}+1)}$, holds iff $\min(e(n_N+1), n_{N+1}+1) = n_N + 1$. As $n_{N+1} > n_N$ gives $n_{N+1} + 1 > n_N + 1$ always, this holds iff $e(n_N+1) = n_N + 1$, that is (as $n_N + 1 \geq 1$) iff $e = 1$.

So if $e \neq 1$, the recursion fails at every N, for every choice of $(n_N)_N$: ϕ is not taut. If $e = 1$: $\phi(\mathfrak{m}^{n+1}) = \mathfrak{n}^{n+1}$ for every $n \geq 0$, so the recursion holds automatically, for every N and every choice of $n_{N+1} > n_N$; so ϕ is taut iff $B = \phi(A) + \mathfrak{n}^{n_0+1}$ for some $n_0 \geq 0$ – equivalently (as $\mathfrak{n}^{n_0+1} \subseteq \mathfrak{n}$ makes this weakest, hence necessary and sufficient, at $n_0 := 0$) iff $B = \phi(A) + \mathfrak{n}$. ∎

Lemma 3.33.11 (Functoriality of Completion, DVR Case)
*Let $(A, \mathfrak{m})$, $(B, \mathfrak{n})$ be DVRs, with completions $\big(\hat{A}, \hat{\mathfrak{m}}\big)$, ι_A, λ_n^A and $\big(\hat{B}, \hat{\mathfrak{n}}\big)$, ι_B, λ_n^B (3.33.4), and $\phi : A \to B$ a local ring homomorphism with ramification index e (3.30.10). This is (3.13.27) (ϕ being continuous (3.33.9)), giving: there is a unique continuous ring homomorphism $\hat{\phi} : \hat{A} \to \hat{B}$ with $\hat{\phi} \circ \iota_A = \iota_B \circ \phi$, which is moreover **local** (3.33.9).*

Moreover this is functorial: $\widehat{\mathrm{id}_A} = \mathrm{id}_{\hat{A}}$; and, for a further DVR $(C, \mathfrak{a}_C)$ and a local ring homomorphism $\psi : B \to C$ (so $\psi \circ \phi : A \to C$ is again local), $\widehat{\psi \circ \phi} = \hat{\psi} \circ \hat{\phi} : \hat{A} \to \hat{C}$.

Moreover $\hat{\phi}$ has the same ramification index e; so by (3.30.10) a), $\hat{\phi}$ is injective if and only if $e < \infty$.

Moreover $\hat{\phi}$ is bijective if and only if it is surjective, if and only if $e = 1$ and $B = \phi(A) + \mathfrak{n}$.

Proof. Drafted by Claude Sonnet, reviewed by David Rufino.

By (3.33.9), ϕ is continuous; so (3.13.27) gives the stated $\hat{\phi}$ and functoriality directly.

As $\hat{A}, \hat{B}$ are DVRs (3.33.8) with maximal ideals $\hat{\mathfrak{m}}, \hat{\mathfrak{n}}$, and $\hat{\phi}$ is continuous for exactly their native filtrations (these being $\ker \lambda_\bullet^A, \ker \lambda_\bullet^B$), (3.33.9) (applied to $\hat{A}, \hat{B}, \hat{\phi}$) gives $\hat{\phi}$ local; and this is the unique local map, uniqueness among local maps coinciding with uniqueness among continuous ones (by that lemma).

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Ramification index. Fix a uniformiser π of A, so $e = v_B(\phi(\pi))$, and $\iota_A(\pi)$ is a uniformiser of $\hat{A}$ (3.33.8) a). For $b \in B$, $b \neq 0$: $\iota_B(b)$ has λ_m^B-coordinate $b \mod \mathfrak{n}^{m+1}$ for every m (3.13.16), so, by (3.33.8) b), $v_{\hat{B}}(\iota_B(b)) = \min\{m \geq 0 : b \notin \mathfrak{n}^{m+1}\} = v_B(b)$. So, by the commuting square,

$$v_{\hat{B}}\big(\hat{\phi}(\iota_A(\pi))\big) = v_{\hat{B}}(\iota_B(\phi(\pi))) = v_B(\phi(\pi)) = e$$

(both sides ∞ if $\phi(\pi) = 0$, by the same computation applied formally): $\hat{\phi}$'s own ramification index (with respect to the uniformiser $\iota_A(\pi)$) is e.

Injectivity criterion. This is (3.30.10) a), applied to $\hat{\phi}$ (with ramification index e, as above).

Surjectivity criterion. If $e = 1$ and $B = \phi(A) + \mathfrak{n}$: by (3.33.10), ϕ is taut, so (3.13.27)'s own surjectivity criterion gives $\hat{\phi}$ surjective. Conversely, suppose $\hat{\phi}$ is surjective; by (3.30.10) b) (applied to $\hat{\phi}$, ramification index e), $e = 1$, so $\phi(\pi)$ is a uniformiser of B (3.30.2), giving $\phi(\mathfrak{m}^{n+1}) = (\phi(\pi)^{n+1}) = \mathfrak{n}^{n+1}$ for every $n \geq 0$; so (3.3.56) gives induced maps $\overline{\phi}_n : A/\mathfrak{m}^{n+1} \to B/\mathfrak{n}^{n+1}$ for every $n \geq 0$, and $\hat{\phi} = \varprojlim \overline{\phi}_n$ (3.3.156) (taking $n(m) := m$ throughout), that is $\lambda_n^B \circ \hat{\phi} = \overline{\phi}_n \circ \lambda_n^A$ for every $n \geq 0$. As λ_n^A, λ_n^B are surjective (3.13.24) and $\hat{\phi}$ is surjective, $\overline{\phi}_n \circ \lambda_n^A$ is surjective, hence so is $\overline{\phi}_n$ (its image already equals that of $\overline{\phi}_n \circ \lambda_n^A$, λ_n^A being surjective); taking $n := 0$, $\overline{\phi}_0$ surjective is exactly $B = \phi(A) + \mathfrak{n}$. Finally, bijective trivially gives surjective, and conversely (3.30.10) b) (applied to $\hat{\phi}$) already gives that $\hat{\phi}$ surjective implies $\hat{\phi}$ bijective: so $\hat{\phi}$ is bijective if and only if it is surjective. ∎

Proposition 3.33.12 (Universal Property of Completion, DVR Case)
*Let $(A, \mathfrak{m})$ be a DVR and $\big(\hat{A}, \hat{\mathfrak{m}}\big)$ its completion (3.33.4), $(B, \mathfrak{n})$ a complete DVR, with $\iota_B : B \xrightarrow{\sim} \hat{B}$ its own completion (3.33.5), and $\phi : A \to B$ a local homomorphism with ramification index e (3.30.10). This is (3.13.28) (ϕ being continuous (3.33.9)), giving: there is a unique **local** ring homomorphism $\hat{\phi} : \hat{A} \to B$ such that $\hat{\phi} \circ \iota = \phi$ (local and continuous being equivalent here (3.33.9)).*

Moreover $\hat{\phi}$ is injective if and only if $e < \infty$.

Further $\hat{\phi}$ is bijective if and only if it is surjective, if and only if $e = 1$ and $B = \phi(A) + \mathfrak{n}$.

Proof. Drafted by Claude Sonnet, reviewed by David Rufino.

By (3.33.9), ϕ is continuous, with $\phi(\mathfrak{m}^{n+1}) \subseteq \mathfrak{n}^{n+1}$ for every $n \geq 0$; and B is complete (3.13.20) (ι_B being an isomorphism). So (3.13.28) gives the stated $\hat{\phi}$ (for the filtration $\ker \lambda_\bullet = \hat{\mathfrak{m}}^{\bullet+1}$ of $\hat{A}$, (3.33.8) e)), namely $\hat{\phi} = \iota_B^{-1} \circ \Psi$ where $\Psi : \hat{A} \to \hat{B}$ is the unique continuous ring homomorphism with $\Psi \circ \iota = \iota_B \circ \phi$ – exactly (3.33.11)'s own $\hat{\phi}$ for ϕ. As ι_B is an isomorphism, $\hat{\phi}$ is injective (resp. surjective) if and only if Ψ is.

As $\hat{A}, B$ are DVRs (3.33.8) (applied to A), with maximal ideals $\hat{\mathfrak{m}}, \mathfrak{n}$, and $\hat{\phi}$ is continuous for exactly their native filtrations $\hat{\mathfrak{m}}^{\bullet+1}$, $\mathfrak{n}^{\bullet+1}$, (3.33.9) (applied to $\hat{A}, B, \hat{\phi}$) gives $\hat{\phi}$ local; and this is the unique local map making the triangle commute, uniqueness among local maps coinciding with uniqueness among continuous ones (3.33.9).

Injectivity criterion. By (3.33.11)'s own injectivity criterion, Ψ is injective if and only if $e < \infty$; so, by the above, so is $\hat{\phi}$.

Surjectivity criterion. By (3.33.11)'s own surjectivity criterion, Ψ is bijective if and only if it is surjective, if and only if $e = 1$ and $B = \phi(A) + \mathfrak{n}$; so, by the above ($\hat{\phi} = \iota_B^{-1} \circ \Psi$, an isomorphism composed with Ψ), the same holds for $\hat{\phi}$. ∎

Proposition 3.33.13 (The Completion Matches the Filtration)
Let $(A, \mathfrak{m})$ be a DVR and $\big(\hat{A}, \hat{\mathfrak{m}}\big)$ its completion (3.33.8). There is a unique ring homomorphism $\nu_n : A/\mathfrak{m}^{n+1} \to \hat{A}/\hat{\mathfrak{m}}^{n+1}$ such that the square

$$\begin{array}{ccc} A & \xrightarrow{\ \iota\ } & \hat{A} \\ {\scriptstyle \overline{\pi}_n}\downarrow & & \downarrow{\scriptstyle \pi_n} \\ A/\mathfrak{m}^{n+1} & \xrightarrow[\ \nu_n\ ]{} & \hat{A}/\hat{\mathfrak{m}}^{n+1} \end{array}$$

commutes. Further ν_n is an isomorphism, with inverse $\mu_n := \nu_n^{-1}$. Moreover, writing $\rho_n : A/\mathfrak{m}^{n+1} \to A/\mathfrak{m}^n$ and $\hat{\rho}_n : \hat{A}/\hat{\mathfrak{m}}^{n+1} \to \hat{A}/\hat{\mathfrak{m}}^n$ for the natural reductions (3.3.56) $(\nu_n)_{n\geq 0}$ is natural in n: for every $n \geq 1$, the square

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$$\begin{array}{ccc} A/\mathfrak{m}^{n+1} & \xrightarrow[\sim]{\nu_n} & \hat{A}/\hat{\mathfrak{m}}^{n+1} \\ \rho_n \downarrow & & \downarrow \hat{\rho}_n \\ A/\mathfrak{m}^{n} & \xrightarrow[\sim]{\nu_{n-1}} & \hat{A}/\hat{\mathfrak{m}}^{n} \end{array}$$

commutes, that is $\hat{\rho}_n \circ \nu_n = \nu_{n-1} \circ \rho_n$ *(equivalently,* $\rho_n \circ \mu_n = \mu_{n-1} \circ \hat{\rho}_n$*).*

Proof. Drafted by Claude Sonnet, reviewed by David Rufino.

Writing $\lambda_n : \hat{A} \to A/\mathfrak{m}^{n+1}$ for the projections (3.33.4): $\ker \lambda_n = \hat{\mathfrak{m}}^{n+1}$ (3.33.8) e), so the statement is exactly (3.13.30) (with $\mathfrak{a} := \mathfrak{m}$), applied using this equality in place of the general inclusion $\mathfrak{m}^{n+1}\hat{A} \subseteq \ker \lambda_n$. ∎

Corollary 3.33.14 (The Completion is Complete)
Let $(A, \mathfrak{m})$ *be a DVR, and* $\hat{A}$ *as in (3.33.4). Then* $\hat{A}$ *is complete (3.33.5).*

Proof. Drafted by Claude Sonnet, reviewed by David Rufino.

By (3.13.26) (applied to $(A, \mathfrak{a}_\bullet)$, $\mathfrak{a}_n := \mathfrak{m}^{n+1}$ (3.13.19)), $\left(\hat{A}, \ker \lambda_\bullet\right)$ is complete (3.13.20) as $\ker \lambda_n = \hat{\mathfrak{m}}^{n+1}$ for every $n \geq 0$ (3.33.8) e), this is exactly completeness of $\hat{A}$ in the sense of (3.33.5). ∎

Lemma 3.33.15 (Completeness of the Fraction Field)
Let $(A, \mathfrak{m})$ *be a complete DVR,* π *a uniformiser, and* $K := \mathrm{Frac}(A)$*. Then* $(\pi^n A)_{n\in\mathbb{Z}}$ *is a filtration of* K *as an* A*-module (3.13.2) (each* $\pi^n A$ *free of rank* 1*), with* $K = \bigcup_n \pi^n A$*; and the canonical map*

$$\jmath : K \to \varprojlim_{n\geq 0} K/\pi^n A, \quad x \mapsto (x \mod \pi^n A)_{n\geq 0},$$

(3.13.17) is bijective.

Proof. Drafted by Claude Sonnet, reviewed by David Rufino.

Filtration, exhaustiveness. $\pi^{n+1}A \subseteq \pi^n A$ for every $n \in \mathbb{Z}$ (as $\pi^{n+1} = \pi \cdot \pi^n$), and $x \mapsto \pi^n x$ is an A-module isomorphism $A \xrightarrow{\sim} \pi^n A$: a filtration of K by finite free A-submodules (3.13.2). Every nonzero $x \in K$ is $x = u\pi^{v(x)}$ for a unit $u \in A^\star$ and its valuation $v(x) \in \mathbb{Z}$ (3.30.2), so $x \in \pi^{v(x)}A$: $K = \bigcup_n \pi^n A$.

Surjectivity of $\jmath$. Let $(\overline{x}_n)_{n\geq 0}$, $\overline{x}_n \in K/\pi^n A$, be compatible under reduction. Pick a representative $x_0 \in K$ of $\overline{x}_0$ and, by exhaustiveness, $N \geq 1$ with $x_0 \in \pi^{-N}A$. Choosing representatives $x_n \in K$ of $\overline{x}_n$ $(n \geq 0)$ inductively with $x_n - x_{n-1} \in \pi^{n-1}A$ for $n \geq 1$ (possible by compatibility): as $\pi^{n-1}A \subseteq \pi^0 A \subseteq \pi^{-N}A$ for every $n \geq 1$, every x_n lies in $M := \pi^{-N}A$.

As $\mathfrak{m}^{n+N}M = \pi^{n+N}A \cdot \pi^{-N}A = \pi^n A$ for every $n \geq 0$, the sequence $(x_n \mod \pi^n A)_{n\geq 0}$, viewed inside M, matches $(x_n \mod \mathfrak{m}^{n+N}M)_{n\geq 0}$. Extend it to a compatible sequence $(\overline{y}_j)_{j\geq 1}$, $\overline{y}_j \in M/\mathfrak{m}^j M$, by setting $\overline{y}_{n+N} := x_n \mod \mathfrak{m}^{n+N}M$ for $n \geq 0$ (compatible among themselves, as the x_n already are) and, for $1 \leq j < N$, $\overline{y}_j$ the image of $\overline{y}_N = x_0 \mod \mathfrak{m}^N M$ under the reduction map $M/\mathfrak{m}^N M \to M/\mathfrak{m}^j M$ (so compatible with the rest too). By (3.33.7) (A complete, M free of rank 1), $M \to \varprojlim_{j\geq 1} M/\mathfrak{m}^j M$ is bijective, so there is $x \in M$ with $x \equiv \overline{y}_j \pmod{\mathfrak{m}^j M}$ for every $j \geq 1$; in particular, at $j = n + N$ $(n \geq 0)$, $x \equiv x_n \equiv \overline{x}_n \pmod{\pi^n A}$. So $\jmath(x) = (\overline{x}_n)_n$: $\jmath$ is surjective.

Bijectivity. $\bigcap_n \pi^n A = \bigcap_{n\geq 0} \mathfrak{m}^n = 0$ (3.30.9), so $(K, \pi^\bullet A)$ is separated; by (3.13.20), surjectivity of $\jmath$ then gives bijectivity. ∎

Proposition 3.33.16 (Completion of the Fraction Field)
Let $(A, \mathfrak{m})$ *be a DVR,* π *a uniformiser,* $K := \mathrm{Frac}(A)$*, and* $\hat{A}, \iota, \lambda_\bullet$ *its completion (3.33.8),* $K_{\mathfrak{m}} := \mathrm{Frac}\left(\hat{A}\right)$*. Write* $\hat{K}$*,* $\jmath$ *for the completion of* K *as a filtered* A*-module for* $(\pi^n A)_{n\in\mathbb{Z}}$ *(3.13.17) (3.33.15). Then* ι *extends (uniquely, (3.5.5)) to an injective ring homomorphism* $\iota_K : K \to K_{\mathfrak{m}}$*, and there is a unique ring structure on* $\hat{K}$ *for which the canonical bijection* $\Psi : \hat{K} \xrightarrow{\sim} K_{\mathfrak{m}}$ *(constructed below) is a ring isomorphism, with* $\Psi \circ \jmath = \iota_K$*.*

Proof. Drafted by Claude Sonnet, reviewed by David Rufino.

As $\bigcap_n \mathfrak{m}^n = 0$ (3.30.9), ι is injective, so (3.5.5) extends it to an injective ring homomorphism $\iota_K : K \to K_{\mathfrak{m}}$, $\iota_K(a/s) = \iota(a)/\iota(s)$. Write $\hat{\pi} := \iota(\pi)$, a uniformiser of $\hat{A}$ (3.33.8).

Exactness. For $a \in A$, $j \geq 0$: $\ker \lambda_{j-1} = \hat{\mathfrak{m}}^j = \hat{\pi}^j \hat{A}$ (3.33.8) (reading $\ker \lambda_{-1} := \hat{A}$ for $j = 0$, trivially true), and $\lambda_{j-1} \circ \iota$ is the reduction map $A \to A/\mathfrak{m}^j$, so $\iota(a) \in \hat{\pi}^j \hat{A} \iff a \in \mathfrak{m}^j = \pi^j A$. For general $a/s \in K$ $(a, s \in A,\ s \neq 0)$, write $s = u\pi^{v(s)}$ for a unit $u \in A^\star$ (so $\iota(u) \in \hat{A}^\star$): fixing $n \geq 0$, $\hat{\pi}^n \iota(s)\hat{A} = \hat{\pi}^{n+v(s)}\hat{A}$ ($\iota(u)$ a unit), so

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$$\iota_K(a/s) \in \hat{\pi}^n \hat{A} \iff \iota(a) \in \hat{\pi}^{n+v(s)} \hat{A} \iff a \in \pi^{n+v(s)} A \iff a/s \in \pi^n A.$$

So $\iota_K^{-1}\left(\hat{\pi}^n \hat{A}\right) = \pi^n A$ for every $n \geq 0$.

Density. Given $y \in K_{\mathfrak{m}}$, $n \geq 0$: $y \in \hat{\pi}^{-M} \hat{A}$ for some $M \geq 0$ (every nonzero element of $K_{\mathfrak{m}}$ has a valuation, (3.33.8)), say $y = \hat{\pi}^{-M} \hat{a}$, $\hat{a} \in \hat{A}$; as $\lambda_{n+M} \circ \iota$ is the (surjective) reduction map $A \to A/\mathfrak{m}^{n+M+1}$, $\hat{a} = \iota(a) + c$ for some $a \in A$, $c \in \ker \lambda_{n+M} = \hat{\mathfrak{m}}^{n+M+1}$; so $y = \iota_K(\pi^{-M} a) + \hat{\pi}^{-M} c$, with $\hat{\pi}^{-M} c \in \hat{\pi}^{-M} \hat{\mathfrak{m}}^{n+M+1} = \hat{\pi}^{n+1} \hat{A} \subseteq \hat{\pi}^n \hat{A}$. So $K_{\mathfrak{m}} = \iota_K(K) + \hat{\pi}^n \hat{A}$.

The bijection Ψ. By Exactness and Density, ι_K induces, for every $n \geq 0$, a bijection $K/\pi^n A \xrightarrow{\sim} K_{\mathfrak{m}}/\hat{\pi}^n \hat{A}$, compatible with reduction (both induced by ι_K); assembling over n, $\hat{K} = \varprojlim_n (K/\pi^n A) \xrightarrow{\sim} \varprojlim_n \left(K_{\mathfrak{m}}/\hat{\pi}^n \hat{A}\right)$. As $\hat{A}$ is complete (3.33.14), (3.33.15) (applied to the complete DVR $\hat{A}$, uniformiser $\hat{\pi}$) gives that $K_{\mathfrak{m}} \to \varprojlim_n \left(K_{\mathfrak{m}}/\hat{\pi}^n \hat{A}\right)$ is bijective; composing with its inverse gives $\Psi : \hat{K} \xrightarrow{\sim} K_{\mathfrak{m}}$. Unwinding the construction on $\jmath(x)$ for $x \in K$ gives $\Psi(\jmath(x)) = \iota_K(x)$: $\Psi \circ \jmath = \iota_K$.

Transport. Define $x \cdot_{\hat{K}} y := \Psi^{-1}(\Psi(x)\Psi(y))$ for $x, y \in \hat{K}$: this makes $\hat{K}$ a ring and Ψ a ring isomorphism, by construction. As $\Psi \circ \jmath = \iota_K$ is then a ring homomorphism and Ψ a ring isomorphism, $\jmath = \Psi^{-1} \circ \iota_K$ is a ring homomorphism too. Uniqueness of a ring structure on $\hat{K}$ making Ψ an isomorphism is immediate, a ring structure transported along a bijection being unique. ∎

Remark 3.33.17

*So, up to the ring isomorphism Ψ, the completion of $K = \mathrm{Frac}(A)$ as a filtered A-module for the fractional-ideal filtration $(\pi^n A)_{n \in \mathbb{Z}}$ – **not** the filtration induced from A (3.13.2), which is degenerate ($\mathfrak{a}_n K = K$ already for $n \geq 0$, K being a field) – recovers exactly $K_{\mathfrak{m}} = \mathrm{Frac}\left(\hat{A}\right)$, with $\jmath$ corresponding to the canonical field extension ι_K.*

Corollary 3.33.18 (The Completion via any Exponent)

Let $(A, \mathfrak{m})$ be a DVR, and $\hat{A}$ as in (3.33.4). For every integer $e \geq 1$ there is a canonical isomorphism $\theta_{\mathfrak{m},e} : \hat{A} \xrightarrow{\sim} \varprojlim_N A/\mathfrak{m}^{Ne}$, $(\overline{a}_n)_{n \geq 0} \mapsto (\overline{a}_{Ne-1})_{N \geq 1}$.

Proof. Drafted by Claude Sonnet, reviewed by David Rufino.

This is (3.13.31) with $\mathfrak{a} := \mathfrak{m}$, noting $\hat{A}_{\mathfrak{m}} = \hat{A}$. ∎

Proposition 3.33.19 (Uniqueness of the Prime Above a Complete Base)

Let $(\mathcal{O}_K, \mathfrak{m})$ be a complete DVR, $K := \mathrm{Frac}(\mathcal{O}_K)$, L/K a finite separable extension, and $\mathcal{O}_L$ the integral closure of $\mathcal{O}_K$ in L. Then $\mathcal{O}_L$ has a unique prime over $\mathfrak{m}$.

Proof. Drafted by Claude Sonnet, reviewed by David Rufino.

As $\mathcal{O}_K$ is a PID and L/K is finite separable, (3.31.11) gives that $\mathcal{O}_L$ is a Dedekind domain, finite free of rank $n := [L : K]$ over $\mathcal{O}_K$. Write $\mathfrak{q}_1, \ldots, \mathfrak{q}_s$ for the primes of $\mathcal{O}_L$ over $\mathfrak{m}$ (there is at least one, by Lying Over (3.23.18), and – $\mathcal{O}_L$ being Dedekind and integral over the DVR $\mathcal{O}_K$ – these are exactly the maximal ideals of $\mathcal{O}_L$), with ramification indices e_l; by the Ramification Formula (3.31.20), $\mathfrak{m}\mathcal{O}_L = \prod_l \mathfrak{q}_l^{e_l}$.

As $\mathcal{O}_K$ is complete and $\mathcal{O}_L$ is finite free over it, (3.33.7) gives that $\mathcal{O}_L \to \varprojlim_N \mathcal{O}_L/\mathfrak{m}^N \mathcal{O}_L$ is bijective: $\mathcal{O}_L$ is $\mathfrak{m}\mathcal{O}_L$-adically complete (3.13.21).

The ideals $\mathfrak{q}_l^{e_l}$ ($l = 1, \ldots, s$) are pairwise comaximal, so the Chinese Remainder Theorem (3.3.63) gives a ring isomorphism $\mathcal{O}_L/\mathfrak{m}\mathcal{O}_L \xrightarrow{\sim} \prod_{l=1}^s \mathcal{O}_L/\mathfrak{q}_l^{e_l}$. If $s \geq 2$, the right side has a nontrivial idempotent (e.g. $(1, 0, \ldots, 0)$, not 0 or 1 as $s \geq 2$); by (3.13.36) (applied to $\mathcal{O}_L$, $\mathfrak{m}\mathcal{O}_L$, complete above), this lifts to an idempotent $e \in \mathcal{O}_L$ with $e \not\equiv 0, 1 \pmod{\mathfrak{m}\mathcal{O}_L}$, so $e \neq 0, 1$ in $\mathcal{O}_L$. But $\mathcal{O}_L$, a subring of the field L, is a domain, so $e^2 = e$ forces $e(e-1) = 0$, hence $e = 0$ or $e = 1$: a contradiction. So $s = 1$. ∎

Corollary 3.33.20 (Uniqueness of the Extended Valuation)

*Let $(\mathcal{O}_K, \mathfrak{m})$ be a complete DVR, $K := \mathrm{Frac}(\mathcal{O}_K)$, $k := \mathcal{O}_K/\mathfrak{m}$, and L/K a finite separable extension of degree n. Let $\mathcal{O}_L$ be the integral closure of $\mathcal{O}_K$ in L. Then $\mathcal{O}_L$ is itself a **complete** DVR, with a unique prime $\mathfrak{q}$ lying above $\mathfrak{m}$, of ramification index e and residue degree f, with $ef = n$.*

*If moreover $f = 1$ (so L/K is **totally ramified**: $e = n$), then $\mathcal{O}_L = \mathcal{O}_K[\varpi]$ for any uniformiser ϖ of $\mathcal{O}_L$ (3.31.21) and, if $\mathcal{O}_K \cong k[[t]]$ for a uniformiser t, then $\mathcal{O}_L \cong k[[\varpi]]$ and $L \cong k((\varpi))$.*

Proof. Drafted by Claude Sonnet, reviewed by David Rufino.

By (3.33.19), $\mathcal{O}_L$ has a unique prime $\mathfrak{q}$ over $\mathfrak{m}$; writing e, f for its ramification index and residue degree, the Ramification Formula (3.31.20) gives $ef = n$, and $\mathcal{O}_L$, being Dedekind and local (its only nonzero prime being $\mathfrak{q}$), is itself a DVR (3.30.2) g) – that is, a place of L, the unique one over $\mathfrak{m}$.

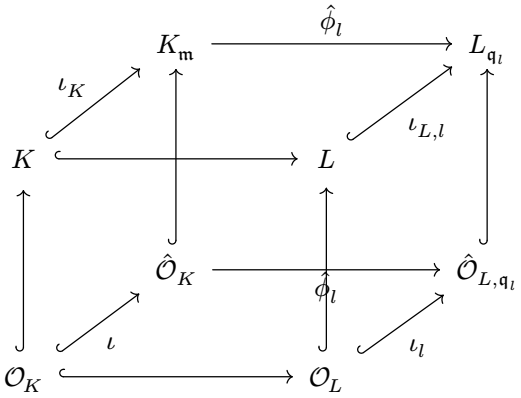
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$\mathcal{O}_L$ is complete. As $\mathcal{O}_K$ is complete and $\mathcal{O}_L$ is finite free of rank n over it (3.31.11), (3.33.7) gives that $\mathcal{O}_L \to \varprojlim_N \mathcal{O}_L/\mathfrak{m}^N\mathcal{O}_L$ is bijective. As $\mathfrak{m}\mathcal{O}_L = \mathfrak{q}^e$ (the unique prime, Ramification Formula), $\mathfrak{m}^N\mathcal{O}_L = \mathfrak{q}^{Ne}$ for every $N \geq 1$; so this says exactly that every sequence compatible under reduction at the levels $\mathfrak{q}^{Ne} = \mathfrak{q}^{n_N+1}$ ($n_N := Ne-1$, $N \geq 1$) arises from an element of $\mathcal{O}_L$. By (3.33.5) (applied to the DVR $\mathcal{O}_L$, this cofinal sequence, c) $\iff$ a)), the same holds for $\mathcal{O}_L$'s own $\mathfrak{q}$-adic tower: $\mathcal{O}_L$ is complete.

If moreover $f = 1$: $e = n$ by $ef = n$. This is exactly the hypothesis of (3.31.21) (a unique prime of $\mathcal{O}_L$ over $\mathfrak{m}$, of residue degree 1), giving $\mathcal{O}_L = \mathcal{O}_K[\varpi]$ for any uniformiser ϖ. Finally, if $\mathcal{O}_K \cong k[[t]]$: $\mathcal{O}_L/\mathfrak{q}$ is a degree-$f = 1$ extension of k, hence equal to k (via the embedding $k \hookrightarrow \mathcal{O}_K \hookrightarrow \mathcal{O}_L \twoheadrightarrow \mathcal{O}_L/\mathfrak{q}$, an isomorphism as $f = 1$): $k \subseteq \mathcal{O}_L$ is a coefficient field. As $\mathcal{O}_L$ is already complete (above), (3.33.3) (applied to $\mathcal{O}_L$, ϖ, k) gives $\mathcal{O}_L \cong k[[\varpi]]$, and $L \cong k((\varpi))$. ∎

Proposition 3.33.21 (Splitting via a Primitive Element)
Let $(\mathcal{O}_K, \mathfrak{m})$ be a DVR, $K := \mathrm{Frac}(\mathcal{O}_K)$, L/K a finite separable extension of degree n, $\mathcal{O}_L$ the integral closure of $\mathcal{O}_K$ in L, with primes $\mathfrak{q}_1, \ldots, \mathfrak{q}_s$ over $\mathfrak{m}$ (Lying Over (3.23.18) these are exactly the maximal ideals of the Dedekind domain $\mathcal{O}_L$ (3.31.11), it being integral over the DVR $\mathcal{O}_K$). Write $\hat{\mathcal{O}}_K, \iota$ for the completion of $\mathcal{O}_K$ (3.33.8), $K_\mathfrak{m} := \mathrm{Frac}\big(\hat{\mathcal{O}}_K\big)$, and $\iota_K : K \to K_\mathfrak{m}$ for the extension of ι (3.5.5) (ι injective, (3.30.9)); similarly, for each l, write $\hat{\mathcal{O}}_{L,\mathfrak{q}_l} := \varprojlim_n \mathcal{O}_L/\mathfrak{q}_l^n$, ι_l, for the completion of $\mathcal{O}_L$ at $\mathfrak{q}_l$ (3.13.16), $L_{\mathfrak{q}_l} := \mathrm{Frac}\big(\hat{\mathcal{O}}_{L,\mathfrak{q}_l}\big)$, and $\iota_{L,l} : L \to L_{\mathfrak{q}_l}$ for the extension of ι_l (ι_l injective, (3.30.9), applied to the DVR $(\mathcal{O}_L)_{\mathfrak{q}_l}$, using $\mathcal{O}_L \cap \mathfrak{q}_l^n(\mathcal{O}_L)_{\mathfrak{q}_l} = \mathfrak{q}_l^n$).

For each l, there is moreover a unique ring homomorphism $\hat{\phi}_l : \hat{\mathcal{O}}_K \to \hat{\mathcal{O}}_{L,\mathfrak{q}_l}$ with $\hat{\phi}_l \circ \iota = \iota_l|_{\mathcal{O}_K}$; it is injective, and extends (uniquely, still denoted $\hat{\phi}_l$) to an injective ring homomorphism $K_\mathfrak{m} \to L_{\mathfrak{q}_l}$, with $\hat{\phi}_l \circ \iota_K = \iota_{L,l}|_K$. These fit into a commutative cube



(the bottom face the rings, the top face their fraction fields, the depth edges the completion maps ι, ι_l, and the vertical edges the canonical inclusions of a domain into its fraction field).

Then there is a canonical isomorphism of $K_\mathfrak{m}$-algebras

$$\Phi : L \otimes_K K_\mathfrak{m} \xrightarrow{\sim} \prod_{l=1}^{s} L_{\mathfrak{q}_l}, \quad x \otimes y \mapsto \big(\iota_{L,l}(x)\hat{\phi}_l(y)\big)_{l=1}^{s}.$$

Proof. Drafted by Claude Sonnet, reviewed by David Rufino.

Structure maps $\hat{\phi}_l$, and the cube. As $\mathfrak{q}_l$ lies over $\mathfrak{m}$, $\iota_l(\mathfrak{m}\mathcal{O}_L) \subseteq \iota_l(\mathfrak{q}_l) \subseteq \hat{\mathfrak{q}}_l$ (continuity of ι_l (3.13.16)): $\iota_l|_{\mathcal{O}_K} : \mathcal{O}_K \to \hat{\mathcal{O}}_{L,\mathfrak{q}_l}$ is local. As $\hat{\mathcal{O}}_{L,\mathfrak{q}_l}$ is a complete DVR, (3.33.12) (applied to $\phi := \iota_l|_{\mathcal{O}_K}$) gives a unique continuous ring homomorphism $\hat{\phi}_l : \hat{\mathcal{O}}_K \to \hat{\mathcal{O}}_{L,\mathfrak{q}_l}$ with $\hat{\phi}_l \circ \iota = \iota_l|_{\mathcal{O}_K}$; it is injective (ramification index $v_{\hat{\mathcal{O}}_{L,\mathfrak{q}_l}}(\iota_l(\pi))$, for a uniformiser π of $\mathcal{O}_K$, finite as $\iota_l(\pi) \neq 0$), so (3.5.5) extends it to an injective ring homomorphism $\hat{\phi}_l : K_\mathfrak{m} \to L_{\mathfrak{q}_l}$; as $\hat{\phi}_l \circ \iota_K$ and $\iota_{L,l}|_K$ agree on $\mathcal{O}_K$ (both restricting to $\iota_l|_{\mathcal{O}_K}$) and $K = \mathrm{Frac}(\mathcal{O}_K)$, the uniqueness clause of (3.5.5) gives $\hat{\phi}_l \circ \iota_K = \iota_{L,l}|_K$ on all of K.

The bottom face of the cube commutes by $\hat{\phi}_l$'s defining property. The left and right faces commute because each diagonal edge ($K \to K_\mathfrak{m}$, $L \to L_{\mathfrak{q}_l}$) is, by construction, the extension along (3.5.5) of the corresponding ring-level map ($\mathcal{O}_K \to \hat{\mathcal{O}}_K$, $\mathcal{O}_L \to \hat{\mathcal{O}}_{L,\mathfrak{q}_l}$). The front face ($\mathcal{O}_K \to \mathcal{O}_L \to L$ versus $\mathcal{O}_K \to K \to L$) is immediate, both being the inclusion $\mathcal{O}_K \to L$. The top face is exactly $\hat{\phi}_l \circ \iota_K = \iota_{L,l}|_K$, just shown.

Primitive element. Fix a primitive element $\theta \in \mathcal{O}_L$ for L/K (3.18.105), with minimal polynomial $g \in \mathcal{O}_K[Y]$ (coefficients integral, as $\mathcal{O}_K$ is integrally closed and θ integral over it), monic of degree n, separable. Viewing $g \in K_\mathfrak{m}[Y]$ via ι_K (still denoted g): as $K_\mathfrak{m}[Y]$ is a UFD, $g = g_1 \cdots g_r$ for monic irreducible $g_j \in K_\mathfrak{m}[Y]$; as g is separable, so is each g_j, and the g_j are pairwise coprime (distinct irreducibles in a PID). As $L \cong K[Y]/(g)$ via $\theta \mapsto Y$ (3.18.40),

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$$L \otimes_K K_{\mathfrak{m}} \cong K_{\mathfrak{m}}[Y]/(g) \cong \prod_{j=1}^{r} K_{\mathfrak{m}}[Y]/(g_j) =: \prod_{j=1}^{r} E_j$$

(the first isomorphism by (3.4.42), the second the Chinese Remainder Theorem (3.3.63) for the pairwise coprime (g_j)). Let $\iota_j : L \to E_j$ be reduction modulo g_j ($\theta \mapsto Y \mod g_j$, well-defined as $g_j \mid g$), a K-algebra map, injective as L is a field; E_j is a field of degree $d_j := \deg g_j$ over $K_{\mathfrak{m}}$, with $\sum_j d_j = n$, and $\iota_j(\theta)$ a root of g_j.

Each factor is a completion. Write $\mathcal{O}_{E_j}$ for the integral closure of $\hat{\mathcal{O}}_K$ in E_j: as $\hat{\mathcal{O}}_K$ is complete and $E_j/K_{\mathfrak{m}}$ is finite separable, (3.33.20) gives that $\mathcal{O}_{E_j}$ is itself a complete DVR, with a unique prime over $\hat{\mathfrak{m}}$. As $E_j = K_{\mathfrak{m}}\big(\iota_j(\theta)\big)$ is the $K_{\mathfrak{m}}$-span of $1, \iota_j(\theta), \dots, \iota_j(\theta)^{d_j-1}$, and $\iota_j(L) = \iota_j(K)\big[\iota_j(\theta)\big]$ the corresponding $\iota_j(K)$-span, while $\iota_j(K)$ is dense in $K_{\mathfrak{m}}$ ($\mathcal{O}_K$ dense in $\hat{\mathcal{O}}_K$ by construction of the completion (3.13.16), and $K, K_{\mathfrak{m}}$ their fraction fields), $\iota_j(L)$ is dense in E_j (coefficientwise, E_j finite-dimensional over $K_{\mathfrak{m}}$). So $\big(\mathcal{O}_{E_j}, \iota_j|_{\mathcal{O}_L}\big)$ is, up to this identification, the completion of $\mathcal{O}_L$ at the prime obtained by pulling back $\mathcal{O}_{E_j}$'s valuation along ι_j: there is a (unique) field isomorphism $\Theta_j : E_j \xrightarrow{\sim} L_{\mathfrak{q}_{l(j)}}$, for some $l(j) \in \{1, \dots, s\}$, with $\Theta_j \circ \iota_j = \iota_{L,l(j)}$ and $\Theta_j|_{K_{\mathfrak{m}}} = \hat{\phi}_{l(j)}$ (both structure maps agreeing, by uniqueness of the extended valuation).

Distinctness. $j \neq j' \implies l(j) \neq l(j')$: equal primes would force $E_j \cong L_{\mathfrak{q}_{l(j)}} = L_{\mathfrak{q}_{l(j')}} \cong E_{j'}$ compatibly with $\iota_j, \iota_{j'}$, hence equal minimal polynomials $g_j = g_{j'}$ for $\iota_j(\theta), \iota_{j'}(\theta)$ over $K_{\mathfrak{m}}$ – contradicting $j \neq j'$ (distinct factors of g).

Exhaustiveness, without a degree count. For each $l \in \{1, \dots, s\}$: as g has $\mathcal{O}_K$-coefficients and $\iota_{L,l}$ is a K-algebra homomorphism, $g\big(\iota_{L,l}(\theta)\big) = \iota_{L,l}(g(\theta)) = 0$; as $\hat{\phi}_l \circ \iota_K = \iota_{L,l}|_K$ (Structure maps, above), this is the same as evaluating the factored $g = g_1 \cdots g_r \in K_{\mathfrak{m}}[Y]$ at $\iota_{L,l}(\theta)$ via $\hat{\phi}_l$, giving $0 = \prod_j g_j\big(\iota_{L,l}(\theta)\big)$ (coefficients of each g_j pushed via $\hat{\phi}_l$), a product of elements of the field $L_{\mathfrak{q}_l}$; so $g_j\big(\iota_{L,l}(\theta)\big) = 0$ for some j, necessarily unique (for $j \neq j'$, a Bézout relation $1 = ag_j + bg_{j'}$ in $K_{\mathfrak{m}}[Y]$, pushed via $\hat{\phi}_l$ and evaluated at $\iota_{L,l}(\theta)$, rules out both vanishing at once) – defining $j(l)$.

We check $j(l(j)) = j$ for every j: by construction of Θ_j, $\Theta_j\big(\iota_j(\theta)\big) = \iota_{L,l(j)}(\theta)$ and $\Theta_j|_{K_{\mathfrak{m}}} = \hat{\phi}_{l(j)}$, so applying Θ_j to $g_j\big(\iota_j(\theta)\big) = 0$ (coefficients via $K_{\mathfrak{m}} \hookrightarrow E_j$) gives $g_j\Big(\iota_{L,l(j)}(\theta)\Big) = 0$ (coefficients via $\hat{\phi}_{l(j)}$); by uniqueness in the definition of $j(\cdot)$, $j(l(j)) = j$. Symmetrically (the roles of $\big(E_j, \iota_j, \hat{\phi}_{l(j)}\big)$ and $\big(L_{\mathfrak{q}_l}, \iota_{L,l}, \hat{\phi}_l\big)$ being interchangeable in this argument, both exhibiting θ's image as a root of the matching factor of g), $l(j(l)) = l$ for every l.

So $j \mapsto l(j)$ and $l \mapsto j(l)$ are mutually inverse bijections between $\{1, \dots, r\}$ and $\{1, \dots, s\}$; in particular $r = s$, and relabelling by $l(j)$,

$$\Phi : L \otimes_K K_{\mathfrak{m}} \xrightarrow{\sim} \prod_{j=1}^{r} E_j \xrightarrow[\prod_j \Theta_j]{\sim} \prod_{l=1}^{s} L_{\mathfrak{q}_l}$$

is an isomorphism of $K_{\mathfrak{m}}$-algebras, with $\Phi(x \otimes 1)$ having l-th coordinate $\Theta_{j(l)}\Big(\iota_{j(l)}(x)\Big) = \iota_{L,l(j(l))}(x) = \iota_{L,l}(x)$ (using $\Theta_{j(l)} \circ \iota_{j(l)} = \iota_{L,l(j(l))}$, above, and $l(j(l)) = l$, just shown) for every $x \in L$. As Φ is $K_{\mathfrak{m}}$-linear and $\Phi(1 \otimes y) = \big(\hat{\phi}_l(y)\big)_l$ for $y \in K_{\mathfrak{m}}$ (the same computation, with θ replaced by the trivial case $x := 1 \in L$, so $\iota_{L,l}(1) = 1$, composed with multiplication by y – equivalently, $1 \otimes y = y \cdot (1 \otimes 1)$ and $\Phi(1 \otimes 1) = (1)_l$), $\Phi(x \otimes y) = \Phi((x \otimes 1)(1 \otimes y)) = \Phi(x \otimes 1)\Phi(1 \otimes y) = \big(\iota_{L,l}(x)\hat{\phi}_l(y)\big)_l$ for every $x \in L$, $y \in K_{\mathfrak{m}}$: the stated formula. ∎

Proposition 3.33.22 (Coefficient Expansion of a Complete DVR)
Let $(A, \mathfrak{m})$ be a DVR with fraction field K and uniformiser π, k a field and A a k-algebra whose residue field $k(\mathfrak{m})$ is separable algebraic over k (for example if k is perfect and $k(\mathfrak{m})/k$ is algebraic), and let $L_\pi : A \to k(\mathfrak{m})[[X]]$ be the injective k-algebra homomorphism of (3.32.17). Then L_π is an isomorphism onto $k(\mathfrak{m})[[X]]$ if and only if A is complete. Moreover there is a unique map $L_{\hat{\pi}} : \hat{A} \to k(\mathfrak{m})[[X]]$ such that $L_{\hat{\pi}} \circ \iota_A = L_\pi$, which is an isomorphism; and when A is complete the extension $L_\pi : K \to k(\mathfrak{m})((X))$ of (3.32.17) is an isomorphism.

Proof. Drafted by Claude Sonnet, reviewed by David Rufino.

$L_{\hat{\pi}}$ exists, is unique, and is bijective. L_π is local: for $a \in \mathfrak{m}$, $c_0(a) = \overline{a} = 0$, so $L_\pi(a) \in (X)$. It is continuous: for $a \in \mathfrak{m}^{n+1}$, $v(a) \geq n+1$, so $v_X(L_\pi(a)) = v(a) \geq n+1$ (or $a = 0$), that is $L_\pi(\mathfrak{m}^{n+1}) \subseteq (X)^{n+1}$ (same-index witness). As $k(\mathfrak{m})[[X]]$ is a complete DVR (3.33.2) (with $t := X$), (3.33.12) (applied to $\phi := L_\pi$) gives a unique continuous ring homomorphism $L_{\hat{\pi}} : \hat{A} \to k(\mathfrak{m})[[X]]$ with $L_{\hat{\pi}} \circ \iota_A = L_\pi$. Its ramification index is $e := v_X(L_\pi(\pi)) = v_X(X) = 1$ (3.30.10), and $k(\mathfrak{m})[[X]] = L_\pi(A) + (X)$ (as the constant term of $L_\pi(a)$ is the residue class $\overline{a} \in k(\mathfrak{m})$ of a, and $A \to A/\mathfrak{m} = k(\mathfrak{m})$ is surjective); so, by that proposition's bijectivity criterion, $L_{\hat{\pi}}$ is bijective.

L_π is an isomorphism onto $k(\mathfrak{m})[[X]]$ iff A is complete. $L_\pi = L_{\hat{\pi}} \circ \iota_A$, with $L_{\hat{\pi}}$ bijective (above); so L_π is bijective if and only if ι_A is. As ι_A is always injective ($\ker \iota_A = \bigcap_n \mathfrak{m}^{n+1} = 0$ (3.30.9)), this holds if and only if ι_A is surjective, that is if and only if A is complete (3.33.5).

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The extension to K. If A is complete, $L_\pi : A \to k(\mathfrak{m})[[X]]$ is moreover surjective (above), so this extension is too: given $y \in k(\mathfrak{m})((X))$, write $y = b/c$ for $b, c \in k(\mathfrak{m})[[X]]$, $c \neq 0$ (3.32.15), so $b = L_\pi(a)$, $c = L_\pi(s)$ for some $a, s \in A$, $s \neq 0$ (as $c \neq 0$, L_π injective), giving $L_\pi(a/s) = y$; so $L_\pi : K \to k(\mathfrak{m})((X))$ is then bijective. Regarding $A \subset \hat{A}$ via ι_A, this commutes with $L_{\hat{\pi}}$: both $L_\pi|_A$ and $L_{\hat{\pi}} \circ \iota_A$ equal the original map $L_\pi : A \to k(\mathfrak{m})[[X]]$ (the latter by construction of $L_{\hat{\pi}}$, above). ■

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